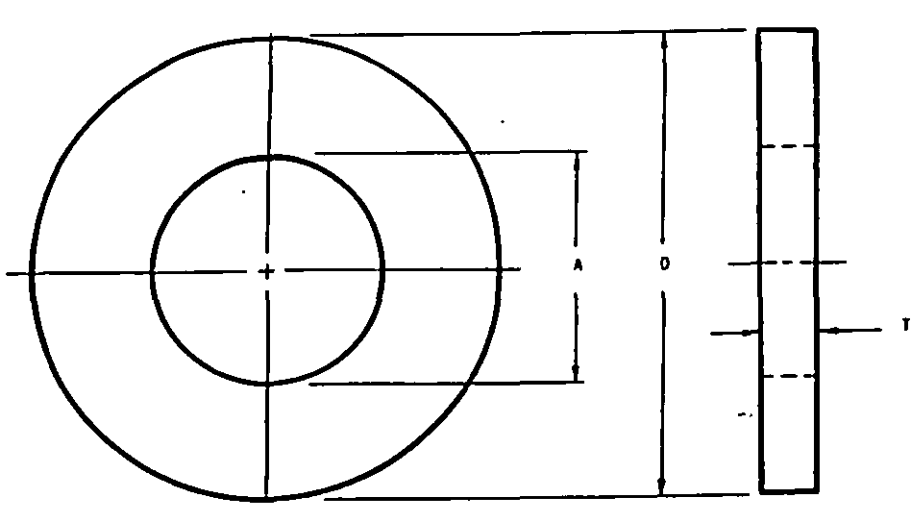
FED. SUP CLASS
5310



REQUIREMENTS:

1. MATERIAL:	CARBON STEEL PER MIL-S-7952	-	NO CODE
	CORROSION RESISTANT STEEL PER MIL-S-5059, AMS 5510 OR AMS 5512.	-	C CODE
	ALUMINUM ALLOY 2024 (ALCLAD) PER QQ-A-250/5, CONDITION T3 OR T4	-	D CODE
	BRASS PER QQ-B-613, COMPOSITION I OR II	-	B CODE
2. COATING:	CADMIUM PLATE PER QQ-P-416, TYPE II, CLASS 2 (FOR CARBON STEEL ONLY)	-	NO CODE
	ANODIZE PER MIL-A-8625	-	K CODE
	CHEMICAL SURFACE TREAT PER MIL-C-5541, CLASS 3	-	J CODE
	BLACK OXIDE COAT PER MIL-C-13924 (FOR CRES ONLY)	-	X CODE
	NATURAL BRASS	-	NO CODE
	PASSIVATE PER QQ-P-35 (FOR UNCOATED CRES ONLY)	-	NO CODE
	ANODIZE PER MIL-A-8625 OR CHEM SURFACE TREAT PER MIL-C-5541, CLASS 1A	-	P CODE

(21) INACTIVE FOR NEW DESIGN. NAS 1149 MAY BE SUBSTITUTED IN ALL APPLICATIONS.

(21) DENOTES CHANGES

APPROVED 18 FEB 42 REVISED (16) 14 DEC 81 (17) 31 JAN 83 (18) 20 JULY 84 (19) 31 JAN 86 (20) 1 APR 86 (21) 06 AUG 92

REVIEWER SYMBOLS:
ARMY - EA, AV
NAVY -
AIR FORCE - 82
DLA - IS

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DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.
NOTE: This drawing was approved by joint action of the Air Force and Navy Departments on the Air Force - Navy standard for this product. This drawing supersedes all associated standard drawings for the same product and shall become effective for the procurement of associated supplies, or for use in new design not later than 6 months after the latest date of approval shown.
AMSC N/A Project No. 5310-1733

P.A. NAVY-AS CUSTODIANS: ARMY-AV NAVY-AS AIR FORCE-99	AIR FORCE-NAVY AERONAUTICAL STANDARD	AN 960
PROCUREMENT SPECIFICATION FF-W-92	WASHER , FLAT	SHEET 1 OF 3

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NOTES:

1. REMOVE ALL BURRS AND SHARP EDGES.

2. DIMENSIONS IN INCHES - TOLERANCES ON THICKNESS:
   FOR ALUMINUM ALLOY WASHER:
   .016 ± .003; .032 ± .004; .042 ± .004; .049 ± .005;
   .063 ∓ .006; .090 ∓ .007.

   ALL OTHER MATERIALS: ± .010.

3. PARALLELISM AND FLATNESS:

   WASHER FACES SHALL BE PARALLEL WITHIN .002 INCH.
   WASHER FACES SHALL BE FLAT WITHIN:
   .007 INCH FOR SIZES UP TO AND INCLUDING THE 1/2 INCH SIZE.
   .010 INCH FOR 9/16 TO 1-1/4 INCH SIZES, INCLUSIVE AND
   .015 INCH FOR 1-5/16 INCH AND LARGER SIZES.

~~4. THICKNESS CODES:~~

~~REGULAR SERIES - NO CODE~~
~~LIGHT SERIES - L CODE~~
~~EXTRA LIGHT SERIES - LL CODE (DESIGNATES .016 THICKNESS WHERE L DESIGNATES .032 THICKNESS)~~
~~MEDIUM SERIES - M CODE (DESIGNATES .032 THICKNESS WHERE L DESIGNATES .016 THICKNESS).~~

5. J CODE WASHERS ARE INTENDED FOR USE WHEN LOW ELECTRICAL RESISTANCE IS REQUIRED.

6. K CODE WASHERS ARE INTENDED FOR USE WHERE CORROSION PROTECTION IS THE PRIME CONSIDERATION.

7. P CODE WASHERS ARE REINSTATED FROM "CANCELLED" TO "INACTIVE FOR NEW DESIGN" AS OF 30 JUNE 1982. USE J OR K CODE WASHERS. P CODE WASHERS MAY BE USED UNTIL STOCK IS DEPLETED.

8. BRASS WASHERS ARE REINSTATED FROM "CANCELLED" TO "INACTIVE FOR NEW DESIGN". AS OF 30 JUNE 1982

9. NO. 2 THREAD SIZES ARE REINSTATED FROM "CANCELLED" TO "INACTIVE FOR NEW DESIGN" AS OF 30 JUNE 1982.

10. EXAMPLE OF PART NUMBER:

AN960 X C 616 LL = CRES STEEL WASHER FOR 3/8 BOLT OR SCREW SIZE, .016 INCH THICK, BLACK OXIDE COATED PER MIL-C-13924

- LL — THICKNESS CODE (SEE ~~NOTE 4~~ AND TABLE I)
- 616 — BOLT OR SCREW SIZE (SEE TABLE I)
- C — MATERIAL CODE (SEE REQ. 1 AND TABLE I)
- X — COATING CODE (SEE REQ. 2 AND TABLE I)
- AN960 — BASIC PART NUMBER

11. MEDIUM SERIES - M CODE CANCELLED AFTER 20 JULY 1984

REV 16 PART NO. DESIGNATION	SUPERSEDING REV 17 PART NO. DESIGNATION
(a) AN960XC10LL AN960JD10LL AN960KD10LL AN960XC10L AN960JD10L AN960KD10L	(b) ~~AN960XC10L~~ ~~AN960JD10L~~ ~~AN960KD10L~~ ~~AN960XC10M~~ ~~AN960JD10M~~ ~~AN960KD10M~~

(a) REINSTATED AFTER 20 JULY 1984

(b) CANCELLED AFTER 20 JULY 1984

APPROVED 18 FEB 4? REVISED (21) FOR CHANGES SEE SHEET 1

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TABLE I. WASHER DIMENSIONS AND DASH NUMBERS

FED. SUP CLASS 5310

BOLT OR SCREW SIZE	A DIA ±.010	B +.070 -.005 DIA	T	DASH NUMBERS							
				CARBON STEEL	CORROSION RESISTANT STEEL		ALUMINUM ALLOY				BRASS
					UNCOATED PASSIVATED	COATED	CHEMICAL CONVERSION COATING	ANODIZED SURFACES	TREATED SURFACES	UNTREATED	
NO. 2	.099	.250	.016 .032	2L 2	C2L C2	XC2L XC2	JD2L JD2	KD2L KD2	PD2L PD2	D2L D2	B2
NO. 3	.105	.250	.016 .032	3L 3	C3L C3	XC3L XC3	JD3L JD3	KD3L KD3	PD3L PD3	D3L D3	B3
NO. 4	.125	.312	.016 .032	4L 4	C4L C4	XC4L XC4	JD4L JD4	KD4L KD4	PD4L PD4	D4L D4	B4
NO. 5	.140	.438	.042	5	C5	XC5	JD5	KD5	PD5	D5	B5
NO. 6	.149	.375	.016 .032	6L 6	C6L C6	XC6L XC6	JD6L JD6	KD6L KD6	PD6L PD6	D6L D6	B6
NO. 8	.174	.375	.016 .032	8L 8	C8L C8	XC8L XC8	JD8L JD8	KD8L KD8	PD8L PD8	D8L D8	B8
NO. 9	.188	.500	.049	9	C9	XC9	JD9	KD9	PD9	D9	B9
NO. 10	.203	.438	.016 .032 .063	10L 10	C10L C10	XC10LL XC10L XC10	JD10LL JD10L JD10	KD10LL KD10L KD10	PD10L PD10	D10L D10	B10
NO. 11	.234	.625	.065	11	C11	XC11	JD11	KD11	PD11	D11	B11
1/4	.265	.500	.016 .032 .063	416L 416	C416L C416	XC416L XC416	JD416L JD416	KD416L KD416	PD416L PD416	D416L D416	B416
5/16	.328	.562	.016 .032 .063	516LL 516L 516	C516LL C516L C516	XC516LL XC516L XC516	JD516LL JD516L JD516	KD516LL KD516L KD516	PD516LL PD516L PD516	D516LL D516L D516	B516
3/8	.390	.625	.016 .032 .063	616LL 616L 616	C616LL C616L C616	XC616LL XC616L XC616	JD616LL JD616L JD616	KD616LL KD616L KD616	PD616LL PD616L PD616	D616LL D616L D616	B616
7/16	.453	.750	.016 .032 .063	716L 716	C716L C716	XC716LL XC716L XC716	JD716LL JD716L JD716	KD716LL KD716L KD716	PD716LL PD716L PD716	D716LL D716L D716	B716
1/2	.515	.875	.016 .032 .063	816L 816	C816L C816	XC816L XC816	JD816L JD816	KD816L KD816	PD816L PD816	D816L D816	B816
9/16	.578	1.062	.016 .032 .063	916L 916	C916L C916	XC916L XC916	JD916L JD916	KD916L KD916	PD916L PD916	D916L D916	B916
5/8	.640	1.188	.016 .032 .063	1016L 1016	C1016L C1016	XC1016L XC1016	JD1016L JD1016	KD1016L KD1016	PD1016L PD1016	D1016L D1016	B1016
3/4	.765	1.312	.016 .032 .090	1216L 1216	C1216L C1216	XC1216L XC1216	JD1216L JD1216	KD1216L KD1216	PD1216L PD1216	D1216L D1216	B1216
7/8	.890	1.500	.016 .032 .090	1416L 1416	C1416L C1416	XC1416L XC1416	JD1416L JD1416	KD1416L KD1416	PD1416L PD1416	D1416L D1416	B1416
1	1.015	1.750	.016 .032 .090	1616L 1616	C1616L C1616	XC1616L XC1616	JD1616L JD1616	KD1616L KD1616	PD1616L PD1616	D1616L D1616	B1616
1-1/16	1.078	1.812	.016 .032 .090	1716L 1716	C1716	XC1716	JD1716L JD1716	KD1716L KD1716	PD1716L PD1716	D1716L D1716	B1716
1-1/8	1.140	1.875	.016 .032 .090	1816L 1816	C1816	XC1816	JD1816L JD1816	KD1816L KD1816	PD1816L PD1816	D1816L D1816	B1816
1-1/4	1.265	2.000	.016 .032 .090	2016L 2016	C2016	XC2016	JD2016L JD2016	KD2016L KD2016	PD2016L PD2016	D2016L D2016	B2016
1-5/16	1.328	2.062	.016 .032 .090	2116L 2116	C2116	XC2116	JD2116L JD2116	KD2116L KD2116	PD2116L PD2116	D2116L D2116	B2116
1-3/8	1.640	2.375	.016 .032 .090	2616L 2516	C2616	XC2616	JD2616L JD2616	KD2616L KD2616	PD2616L PD2616	D2616L D2616	B2616
1-7/8	1.890	2.625	.016 .032 .090	3016L 3016	C3016	XC3016	JD3016L JD3016	KD3016L KD3016	PD3016L PD3016	D3016L D3016	B3016
2-1/4	2.265	3.000	.016 .032 .090	3616L 3616	C3616	XC3616	JD3616L JD3616	KD3616L KD3616	PD3616L PD3616	D3616L D3616	B3616
2-1/2	2.515	3.250	.016 .032 .090	4016L 4016	C4016	XC4016	JD4016L JD4016	KD4016L KD4016	PD4016L PD4016	D4016L D4016	B4016

APPROVED 18 FEB 42 REVISED (21) FOR CHANGES SEE SHEET 1

P.A.:
NAVY-AS
CUSTODIANS:
ARMY-AV
NAVY-AS
AIR FORCE-99
PROCUREMENT SPECIFICATION
FF-W-99

AIR FORCE-NAVY AERONAUTICAL STANDARD

WASHER, FLAT

AN 960

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